

HTHS Final Exam – Algebra 2 Formulas

Combinations: ${}_nC_r = \frac{n!}{(n-r)!r!}$

Permutations: ${}_nP_r = \frac{n!}{(n-r)!}$

Binomial Theorem:

$$(a+b)^n = {}_nC_0 \cdot a^n + {}_nC_1 \cdot a^{n-1}b + {}_nC_2 \cdot a^{n-2}b^2 + \cdots + {}_nC_n \cdot b^n$$

Binomial Experiment:

$$P(k \text{ successes}) = {}_nC_k \cdot p^k(1-p)^{n-k}$$

Arithmetic Sequence/Series:

$$a_n = a_1 + (n-1)d$$

$$S_n = \frac{n(a_1 + a_n)}{2}$$

Geometric Sequence/Series:

$$a_n = a_1r^{n-1}$$

$$S_n = \frac{a_1(1-r^n)}{(1-r)}$$

$$S_\infty = \frac{a_1}{(1-r)} \quad \text{for converging series}$$

Arc Length: $S = r\theta$

Sector Area: $A = \frac{1}{2}\pi r^2\theta$

Law of Sines: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

Law of Cosines: $a^2 = b^2 + c^2 - 2bc \cos A$

Area of a triangle: $A = \frac{1}{2}ab \sin C$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

Tangent and Cotangent Identities:

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities:

$$\sin^2 \theta + \cos^2 \theta = 1 \quad 1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

Cofunction Identities:

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta \quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \quad \tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta$$

Negative Angle Identities:

$$\sin(-\theta) = -\sin \theta \quad \cos(-\theta) = \cos \theta \quad \tan(-\theta) = -\tan \theta$$